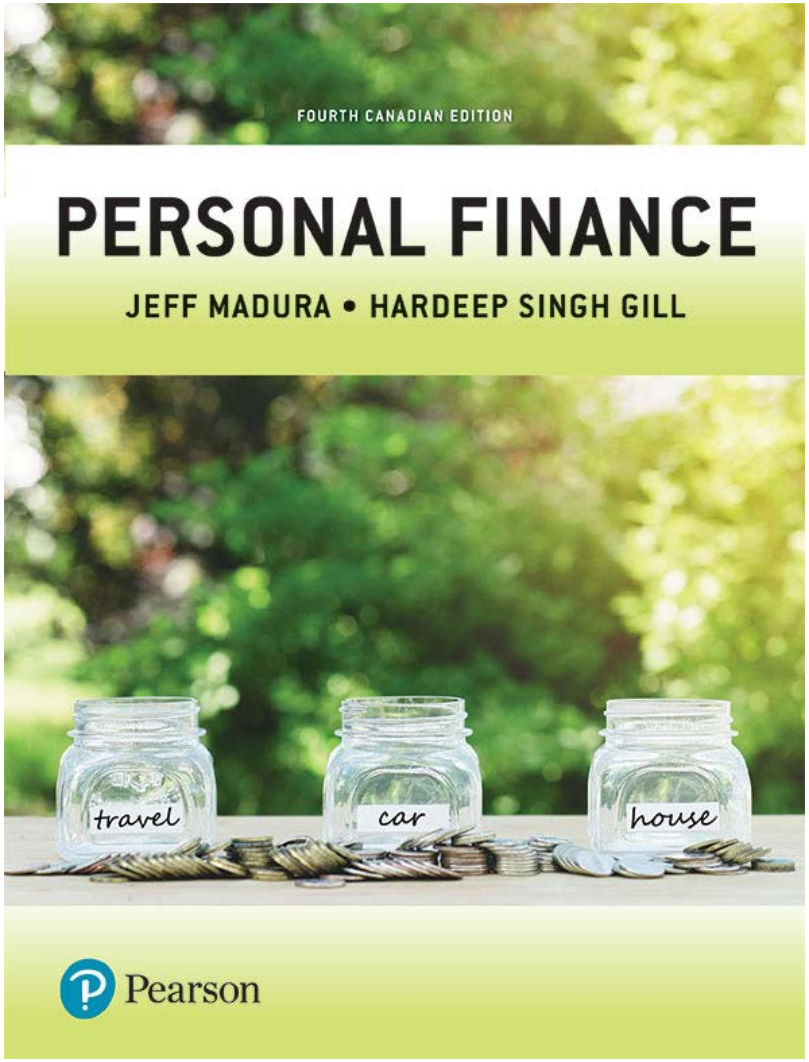
Personal Finance

Fourth Canadian Edition



Chapter 2

Applying Time Value Concepts

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The Importance of the Time Value of Money

- The gain or loss of interest on a dollar amount (when you spend money, you incur an opportunity cost of what you would have done with that money had you not spent it)
- Interest: the rent charged for the use of money (you can pay or receive interest)
- Two ways of computing interest: simple interest and compound interest

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Simple Interest

- interest on a loan computed as a percentage of the loan amount, or principal
  - Interest earned, or paid (and is not reinvested)
  - Measured using the principal (P), the interest rate applied to the principal (r), and the loan's time to maturity (t)

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Simple Interest Formula

$$I = P \times r \times t$$

where

I = interest earned

P = principal, or present value

r = annual interest rate expressed as a decimal or percent

t = time (in years)

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Simple Interest Example

Farah makes a deposit of $1000 in a high-interest savings account paying 3 percent simple interest annually. At the end of year one, the bank will credit Farah's chequing account with $30, calculated as:

$$I = P \times r \times t$$

$$\$30 = \$1000 \times 0.03 \times 1$$

In year two, the initial principal of $1000 will again earn $30 in interest. The $30 will be credited to Farah's chequing account. The table below displays what this process looks like over the first three years of Farah's $1000 investment.

Year	Principal	Interest	Accumulated Interest	Amount Credited to Chequing Account
1	$1000	$30	$30	30
2	1000	30	60	30
3	1000	30	90	30

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Compound Interest

- the process of earning interest on interest
- Ways to solve time value money ('TVM') problems:
  - TVM tables
  - TVM formulas
  - Financial calculator (e.g. TI BA II Plus)

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Lump Sums vs. Annuities

- Time value of money is most commonly applied to two types of cash flows – a single dollar amount (a lump sum) and an annuity
- Annuity: the payment of a series of equal cash flow payments at equal intervals of time

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Compounding Periods Table

- The number of compounding periods per year for different compounding periods:

Compounding Period	Number of Compounding Periods per Year (*n*)
Annually (every year)	1
Semi-annually (every 6 months)	2
Quarterly (every 3 months)	4
Monthly (every month)	12
Weekly (every week)	52
Daily (every day)	365

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Using A Financial Calculator (2 of 2)

- Cash outflows, such as an investment amount, are entered as a negative number
- Cash inflows, such as investment income, are entered as a positive number
- Specify the number of payments per year (P/Y) and the number of compounding periods per year (C/Y) (press 2ND and the key- default is 1 payment/year))

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Two Types of Annuities

- Ordinary annuity: a stream of equal payments that are received, or paid, at equal intervals in time, at the *end* of a period
- Annuity due: a series of equal cash flow payments that occur at the *beginning* of each period

**If the payment changes over time the payment stream does not reflect an annuity

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Using Timelines

- The best way to illustrate the future value of an ordinary annuity or an annuity due is through the use of timelines
- Timelines: diagrams that show payments received or paid over time

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Interest Rate Conversion (1 of 2)

- Nominal interest rate: the stated or quoted, rate of interest
  - Also known as the annual percentage rate (APR)
- When comparing two or more interest rates, the nominal interest rate is *not* useful because it does not take into account the effect of compounding

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Interest Rate Conversion (2 of 2)

- Effective interest rate: the actual rate of interest that you earn, or pay, over a period of time
  - Also known as the effective yield (EY)
- Allows for the comparison of two or more interest rates because it reflects the effect of compound interest

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Interest Rate Conversion Formula

- Convert nominal interest rates to effective interest rates using the following formula:

$$EY = \left[\left(1 + \frac{i}{n} \right)^{n} \right] - 1$$

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Interest Rate Conversion Examples

- Effective yield of 10 percent compounded semi-annually is 10.25%:

$$.1025 = \left[\left(1 + \frac{0.10}{2}\right)^{2}\right] - 1$$

- Effective yield of 10 percent compounded monthly is 10.47%...

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Interest Rate Conversion Table

- Equivalent effective interest rates for various nominal interest rates:

Nominal Interest Rate	Effective Interest Rate
10% compounded annually	10.00%
10% compounded semiannually	10.25%
10% compounded quarterly	10.38%
10% compounded monthly	10.47%
10% compounded weekly	10.51%
10% compounded daily	10.52%

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Effective Yield Using A Financial Calculator

TVM EXAMPLE

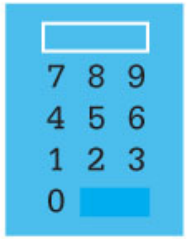
Marco has $4000 invested at 4.5 percent compounded semi-annually at his bank. In order to make a comparison with another financial institution, he needs to know the effective interest rate at his bank. What is the effective annual interest rate? For this problem, the number of compounding periods is two. Use the interest conversion worksheet in the TI BA II Plus calculator to solve as follows:

2ND ICONV 4 · 5 ENTER (the nominal interest rate)

↑ 2 ENTER (number of compounding periods)

↑ CPT (calculate the effective interest rate)

The effective yield is 4.55 percent.



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